

Direct and indirect chemical defence of pine against folivorous insects

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The chemical defence of pine against herbivorous insects has been intensively studied with respect to its effects on the performance and behaviour of the herbivores as well as on the natural enemies of pine herbivores. The huge variety of terpenoid pine components play a major role in mediating numerous specific food web interactions. The constitutive terpenoid pattern can be adjusted to herbivore attack by changes induced by insect feeding or oviposition activity. Recent studies on folivorous pine sawflies have highlighted the role of induced pine responses in herbivore attack and have demonstrated the importance of analysing the variability of pine defence and its finely tuned specificity with respect to the herbivore attacker in a multitrophic context.

Pine entangled in the food web: how to be successful?

If evolutionary success implies adaptive radiation, long-term species survival and wide geographical spread, then pines (*Pinus* spp.) have been successful: they are the largest genus of the conifer family Pinaceae, comprising >100 species and many subspecies and varieties. They are mainly distributed throughout the temperate regions of the northern hemisphere and occupy a wide range of climatically different habitats [1]. They are obliged to cope with a wide range of attackers and have evolved a versatile set of defence mechanisms to do so (Figure 1).

Physical barriers are created by static structures such as lignified cells, calcium oxalate crystals or tough foliage [2]. Chemical defence mechanisms are usually divided into (a) constitutively (preformed) mechanisms that are permanently present and (b) responses that are induced upon herbivore attack. Defence mechanisms can be directed against the behaviour of herbivorous insects by deterring their oviposition or feeding, or against the physiology of the insects by reducing survival or fecundity [3]. In addition, volatile compounds can attract natural enemies of attacking herbivores, such as predators and parasitic wasps, thereby defending plants indirectly against attacking herbivores [4]. Predators and parasitoids can have a significant impact on the population dynamics of forest insects [5]. Plants can influence predation and parasitism upon herbivores in many different ways [6]. Predatory and parasitic insects often use chemical cues from the herbivore and host plant to

find suitable prey or host organisms. In the past few years, conifer defence mechanisms, particularly of stem tissue, and conifer interactions with stem-boring insects have been reviewed in great detail [2,7,8]. However, conifer foliage is also attacked by a wide range of herbivorous insects and mammals, resulting in a considerable amount of damage.

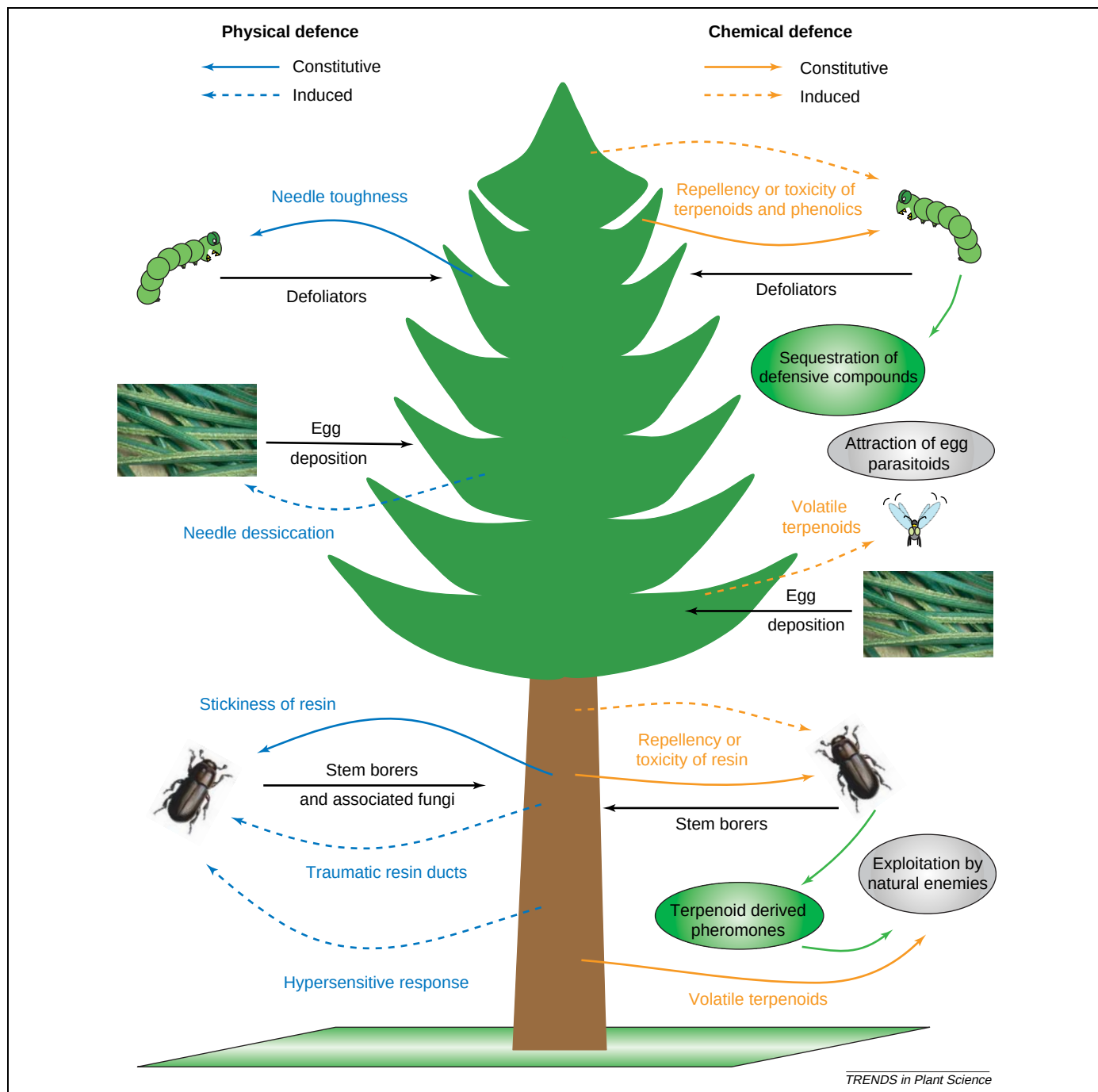
Here we review the chemical defence mechanisms of conifers against defoliating insects, focussing particularly on pines. In addition to asking how pine chemistry affects the performance and behaviour of defoliating herbivores and how some herbivores have evolved adaptations to these defences, we will emphasize the impact of pine chemistry in a multitrophic context and, hence, consider the effect of pine chemistry on the natural enemies of pine herbivores.

Chemical defence of pines against defoliators

Host plant chemistry can strongly influence the preference and performance of defoliating insects even though many herbivores have evolved adaptations to the chemistry of a conifer species [9,10]. The most important defence mechanism of pine and other conifers against folivorous insects is the production of phenolic compounds and oleoresin containing numerous terpenoids. Phenolic compounds are produced by specialized ray cells and polyphenolic parenchyma cells [2]. Large quantities of terpenoid resin are produced by special secretory tissues and stored in a network of specialized resin ducts located throughout the wood, bark, roots and needles [8]. Terpenoid resin is a complex blend comprising a volatile turpentine fraction (monoterpenes, sesquiterpenes) and a less volatile rosin fraction (mainly diterpene resin acids). The biosynthesis of terpenoids has been reviewed extensively and will not be covered here [11,12]. Biochemical and molecular aspects of conifer defences have attracted much attention in recent years. In particular, Norway spruce (*Picea abies*), Sitka spruce (*Picea sitchensis*) and grand fir (*Abies grandis*) have been used as model plants to study the molecular basis of constitutive and induced terpenoid-based defences; other members of the Pinaceae, such as loblolly pine (*Pinus taeda*) and Douglas fir (*Pseudotsuga menziesii*) have also been studied. The molecular biochemistry and genomics of terpenoid-based defences has been covered extensively by recent reviews [13–15].

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Direct defence mechanisms

Conifer resin as a constitutive and inducible defensive device is known to have direct detrimental effects on herbivores as well as indirect ones, by attracting their natural enemies (Table 1). In addition to the physical defensive effects of resin, resin terpenoids are toxic to a broad range of organisms [16]. Terpenoids can usually readily interact with cell membranes, which can lead to uncontrolled cell leakage, finally resulting in cell death. However, surprisingly little is known about the structure–toxicity relationships of terpenoids in conifer-feeding insects. Non-specialized insects are often deterred or killed by resin constituents, whereas the defensive effect

against more adapted insects depends on the resin concentration and composition. Some folivorous sawfly larvae, such as the Diprionidae, have evolved specific morphological structures to use the resin of their host plants. They sequester resin in foregut pouches and regurgitate it upon disturbance. Predators attacking these diprionid larvae are repelled by the resin volatiles and the stickiness of their prey [17,18]. The performance of folivorous insects depends significantly on the host plant species or on different provenances [9,10,19,20].

The role of monoterpenes in the defence of pine and other conifers against defoliators remains controversial. The larval performance and oviposition behaviour of pine

Table 1. Role of terpenoids and phenolics in direct and indirect defence of conifers against herbivores

Affected stage or attacked stage		Chemical plant trait	Effect	Refs
Direct defence^a				
Insect oviposition		Diterpene resin acids	+	[20,25,26]
Egg survival		Water content	–	[21]
Larval performance		Monoterpenes	–	[21,23]
		Sesquiterpenes	0	[22]
		Diterpene resin acids	–	[20,25,26]
		Phenolics	–	[10,20,21]
			0	[10,23,29,31]
Indirect defence^b				
Adult herbivore	Response to host or prey pheromones	Monoterpenes		[35–39]
	Synergism of host or prey pheromones and constitutive plant volatiles			[36,38–42,45–47]
Larvae	Infested bark and fungi	Oxygenated monoterpenes		[49–52]
Eggs	Induced plant volatiles; specific volatile pattern	Sesquiterpene		[55,56,58,59]

^aDirect defence: +, positive effect; –, negative effect; 0, neutral effect.

^bIndirect defence: no effect on the herbivore stage, but positive effect on natural enemies of the herbivores.

sawflies is either not or only slightly negatively correlated with certain monoterpenes, for example, 3-carene [10,20,21]. Furthermore, the resistance of Douglas fir to the western spruce budworm (*Choristoneura occidentalis*) could not be attributed to its ability to biosynthesize great amounts of monoterpenes [22]. However, in experiments in which terpenes were added to artificial diets, several monoterpenes and sesquiterpenes had adverse effects on the growth rates and pupal weight of the western spruce budworm [23].

Several studies have shown a negative relationship between the composition of diterpene resin acids and host tree suitability for defoliating insects [24,25]. Pine sawfly females were observed to lay eggs on needles with pine resin contents that were detrimental to larval performance [20,25,26]. However, even though this behaviour seems maladaptive at first glance, it might enable the offspring to escape high mortality from natural enemies because predation is significantly lower when the larvae feed on pine foliage containing high levels of resin acids [24]. By contrast, sawfly larvae that had fed upon pine foliage with high resin acid concentration were not better protected against parasitism and predation at the cocoon stage [27]. Therefore, producing constitutively high amounts of resin acids is a trade-off for pines, between being well defended directly against conifer sawflies, but reducing the effectiveness of indirect defence mechanisms because of the detrimental effects on some carnivore species.

Phenolic compounds in conifers are considered to be major defence compounds against pathogens and herbivores [2,28]. However, knowledge of the effect of phenolic compounds on defoliating insects on pine as well as of their adaptations to phenolics is limited. Some studies have shown that the larval performance of the diprionid sawfly *Diprion pini* was negatively correlated with the content of the phenolic flavonoid taxifolin [10,20,21], whereas only a weak relationship was found in other studies, [29]. The sawfly *Gilpinia hercyniae*, which feeds upon spruce, was even found to use phenolics for nutrition [30]. Conifer defoliating lepidopteran species also do not seem to be significantly affected by phenolic compounds [10,23,31].

Thus, the evidence for the importance of phenolics for conifer defence is ambiguous.

Often, the resistance or suitability of conifers to defoliators cannot be ascribed to a single class of secondary defensive compounds such as terpenes or phenolics. Moreover, other traits such as budburst phenology, nutrients and needle toughness can mediate or modify the importance of defensive secondary compounds [3,22,32,33]. Furthermore, not only can secondary compounds such as terpenoids have a direct effect on herbivores, but they can also have an indirect effect by providing important chemical cues to natural enemies to guide them to suitable hosts.

Indirect defence mechanisms

Many natural enemies of herbivorous insects use chemical cues to locate suitable host or prey species. These cues can originate from the herbivores, the microorganisms associated with the herbivores, or from the host plants (Table 1). Chemical cues originating directly from the herbivore might be highly reliable for indicating the presence of a suitable host or prey, but relatively difficult to detect because of their minute amounts. By contrast, volatiles emitted from the foliage are emitted in large amounts, particularly in trees with their huge foliage biomass and surface. Thus, odour released from tree foliage might provide highly detectable cues both for herbivores and their natural enemies searching for prey or hosts. However, although highly detectable, the high geno- and phenotypic variation of pine volatiles reduces the reliability of the odour for a foraging predator or parasitoid [34]. Natural enemies of pine-feeding insects have evolved various strategies to overcome this so-called 'reliability–detectability dilemma'.

Many predators and parasitoids are known to be able to detect the highly reliable, species-specific pheromones of their prey or hosts and to exploit them for foraging [35–39]. For example, many predators of bark beetles (Coleoptera, Scolytidae) are attracted by the aggregation pheromones that are emitted by male beetles when colonizing a new host tree [36,38,40–42]. These pheromones are often oxygenated derivatives of plant monoterpenes or can also be synthesized *de novo*, mediated by

scolytid terpene synthases [43]. Many bark beetles also use monoterpenes from the host plant as kairomones either alone or in combination with aggregation pheromones to locate the plants [7]. Benefits of aggregation in bark beetles can include an enhanced attraction of mates, a better chance to overcome the defences of the host plant and reduced risk of predation [44].

The attraction of predators to the reliable bark beetle pheromones can be synergized by plant monoterpenes, which are usually available in highly detectable amounts [36,45,46]. In most cases, monoterpenes alone were not attractive or were less attractive to predators than when combined with pheromones. The synergistic effect of monoterpenes and pheromones can be dose-dependent, although this is strongly influenced by the combination of compounds and predator species [39,41,42,45,47]. For example, the attraction of the predatory beetle *Thanasimus dubius* was positively correlated with the concentration of α -pinene when mixed with ipsdienol and lanerione, the aggregation pheromones of its scolytid prey. By contrast, low and intermediate amounts of α -pinene synergized the attraction of *Ips pini* to its pheromones but higher concentrations reduced the attraction of bark beetles [47,48]. Thus, a high ratio of α -pinene to bark beetle pheromones might be an important component of integrated host defences against certain bark beetles [47].

The synergistic effect of pine terpenoids to bark beetle pheromones can even be enantioselective. For example, (+)- α -pinene synergized the response of *T. dubius* to the pheromone of *Ips pini*, whereas (–)- α -pinene synergized the response to the pheromone of *Ips grandicollis* [36]. Thus, specific enantiomers of plant cues can specify information about the identity of a certain prey species.

The combined use of highly reliable host or prey pheromones with highly detectable plant volatiles is a strategy that predators or parasitoids can use to cope with the reliability–detectability problem when foraging for hosts or prey. This strategy is used particularly by predators and parasitoids that predominantly attack the adult life stage of pine-feeding insects. Parasitoids and predators foraging on later larval instars cannot rely on the presence of aggregation pheromones. Therefore, other chemical cues are used to locate suitable hosts. For example, the larval parasitoid *Roptrocercus xylophagorum* is attracted to the bark of loblolly pine (*Pinus taeda*) infested with bark beetle larvae but not to uninfested or artificially wounded logs [49,50]. Electrophysiological and behavioural studies have revealed that parasitoids particularly respond to oxygenated monoterpenes, but not to most of the predominant monoterpene hydrocarbons such as α -pinene [49–52]. The oxygenated monoterpenes are presumably produced by an interaction between bark beetles, their associated fungi and the host tree phloem, indicating that bark beetles induce changes in the odour bouquet of the host tree, which is amplified by associated fungi [53].

Another strategy to circumvent this problem is to rely on plant cues that are induced after herbivore attack. Many plants defend themselves indirectly against

herbivorous insects by emitting a specific odour blend induced by the attacker. These herbivore-induced plant volatiles can attract predators and parasitoids [4,54]. In addition to the high detectability of these plant cues, they are often highly specific for the herbivore species and even for the insect stage attacking the plant.

Egg deposition by the pine sawfly *D. pini* induces changes in the volatile pattern of *Pinus sylvestris* twigs, rendering the odour of egg-laden twigs attractive to an egg parasitoid that specializes on pine sawfly eggs [55]. The pine response was not only locally restricted to the immediate site of egg deposition but was also systemic (i.e. parts of the same twig that carried no eggs were also attractive to the parasitoid *Chrysonotomyia ruforum*), showing that parasitoids were attracted by induced plant volatiles and not by cues directly related to the host (Figure 2).

Chemical analysis of pine volatiles from oviposition-induced twigs revealed a significantly higher emission of the sesquiterpene (*E*)- β -farnesene than in control twigs [56,57]. Further experiments showed that (*E*)- β -farnesene was able to attract female egg parasitoids when offered in combination with non-induced pine volatiles [58]. Therefore, a minute, but significant quantitative change of the ratio of a single sesquiterpene within the blend of numerous other terpenoid pine components results in the attraction of an egg parasitoid. The egg parasitoid seems to be highly adapted to this finely tuned specific change of volatiles. *P. sylvestris* responds specifically to the egg deposition of the two most abundant sawfly species (*D. pini* and *Neodiprion sertifer*) by releasing volatiles that attract egg parasitoids, whereas pines laden with eggs of a less abundant species (*Gilpinia pallida*) did not emit volatiles attracting the egg parasitoid [59] (Figure 2).

The elicitor responsible for the oviposition-induced emission of pine volatiles attractive to egg parasitoids is located in the oviduct secretion of female *D. pini* sawflies. The oviduct secretion, which coats the eggs, is transferred into the wounded plant tissue during egg deposition. Wounding *per se* does not induce the production of attractive volatiles. But when oviduct secretion is applied to artificially wounded pine needles, the production of volatiles attractive to the egg parasitoids is induced [55,60] (Figure 2). Further studies have indicated that the elicitor within the oviduct secretion is a small protein or a protein-bound component that needs to interact with the disrupted pine needle tissue [60,61]. So far, it is not known whether the response of pine to this elicitor is mediated by a specific receptor or whether the elicitor acts by other means, such as by the formation of ion channels.

The time scale during which induced plant responses to herbivores can occur is variable. The induced effects described here so far can be considered to be so-called rapid induced responses, which directly affect the same generation of the attacking herbivores. Short-term induced responses of secondary plant chemistry have been studied in conifers from the molecular to the ecosystem level [2,13,14,62]. Delayed induced responses in conifers have also been reported (i.e. the response of plants that are exposed to future herbivore generations after having been attacked previously)

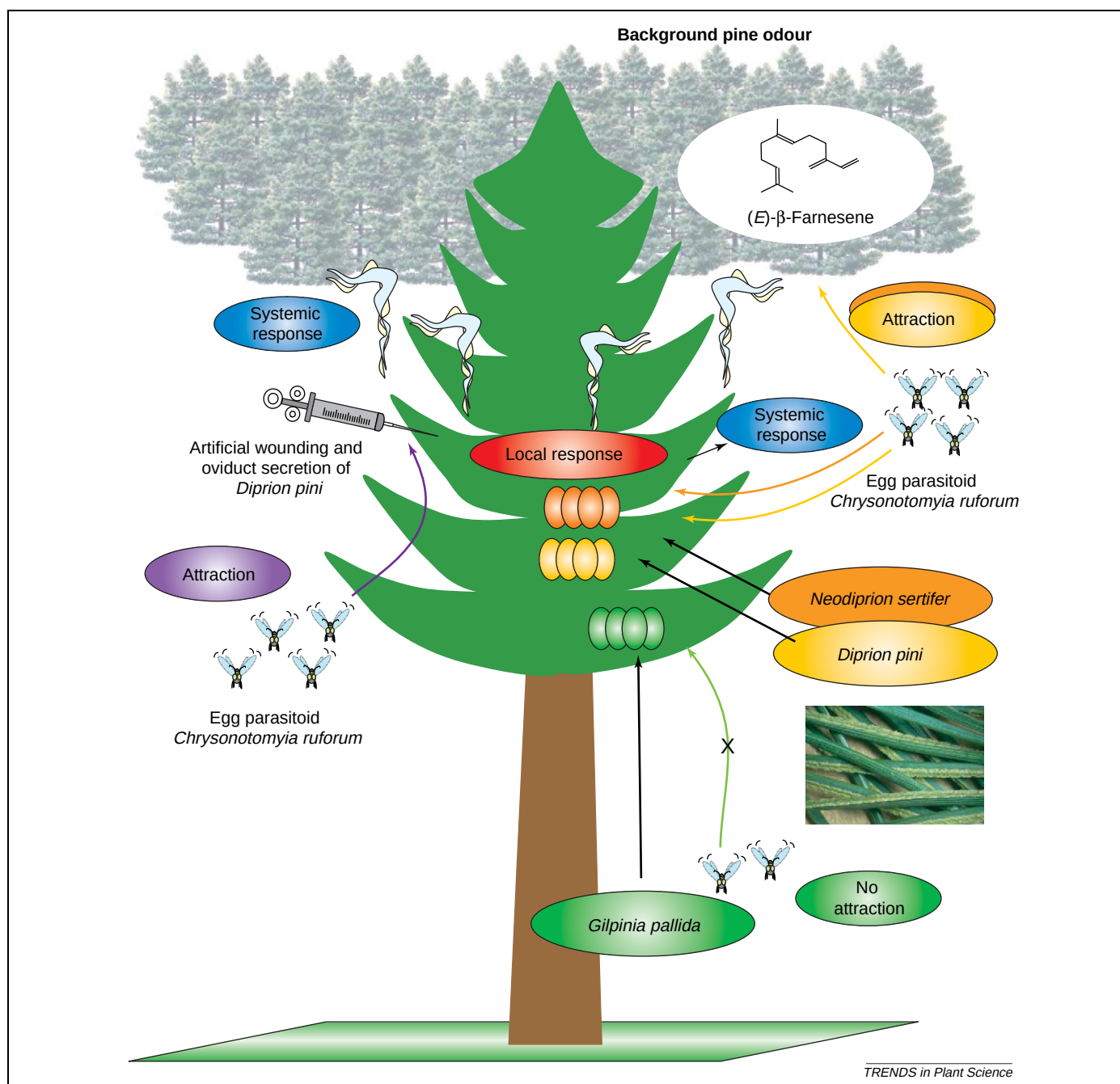


Figure 2. A scenario of the ecological effects of pine responses induced by insect egg deposition.

[63–67]. Some studies showed induced resistance after previous defoliation, whereas others did not [31,64,68,69]. By contrast, needles of previously defoliated trees often tended to be even more suitable for subsequent defoliator generations [70–72]. Whether a previously defoliated tree shows an induced susceptibility or not to defoliators is dependent on many factors, such as tree age, intensity of defoliation, insect species, and the performance parameter of the herbivore that is being investigated [63,73]. Thus, although there might be inducible long-term responses in the foliage of pine and other conifers, these responses do not necessarily result in increased resistance.

When studying the ecological effects of defoliation, most studies have investigated the behaviour and performance of the feeding stages of the defoliating species. However, the performance of less investigated life stages, such as eggs, could be affected as well [3]. In addition, the oviposition behaviour of pine herbivores might also be affected by induced plant responses as has been shown for herbivores of herbaceous plants [74]. Furthermore, the potential effects of long-term induced plant responses on the performance of natural enemies should be considered [75,76]. In addition, defoliation-induced responses are not restricted to the foliage, but can also affect the chemistry in other tissues of the plant, decreasing the resistance to

other herbivore guilds such as stem-boring bark beetles [66,67].

Costs of conifer defence against herbivores

Many members of the Pinaceae produce considerable amounts of defensive chemical compounds. The amount of constitutively produced resin varies greatly within the Pinaceae, with pines having the highest constitutive production of resin, which is stored in the most elaborate network of resin ducts among the pine family [8]. This high level of production requires a considerable amount of metabolic energy because the biosynthesis of terpenoids can be costly [77]. Despite the high metabolic costs, a high constitutive defence level might be favourable when the threat by herbivores is temporally and spatially high. Some members of the pine family are attacked by >200 different insect species, as well as by various birds, mammals and fungi, and are attacked by a greater number of bark beetle species compared with other conifer families [2,78]. Facing the attack of many specialized herbivore species might also cause so-called ecological costs because defensive plant compounds are often exploited by specialists for their own defensive purposes, as outlined already ('Direct defence mechanisms'). Additional ecological costs can emerge when defensive plant compounds adversely affect natural enemies and bottom-up effects counteract a potential top-down control of herbivores [76].

However, inducible defences could minimize the fitness costs of resistance, although some of the costs of being inducible need to be considered [79]. Induced defences can be more specific owing to the interaction of exogenous elicitors and recognition mechanisms by plants. Thus, induced responses might be an adaptive defensive trait that is particularly important in indirect defence mechanisms that are based on reliable signals [80].

However, induced defences have the disadvantage that there is a time gap until the resistance is induced or natural enemies are attracted, which can take days. During this time, plants are still vulnerable to herbivores, which challenges the adaptive advantage of induced defences. The recently demonstrated response of pines to insect egg deposition can be considered a strategy to minimize this crucial time gap: the plants respond before the larvae hatch and before feeding damage by a new herbivore generation occurs [4]. Therefore, such an oviposition-induced plant response has been interpreted as an 'early herbivore alert' [61]. Given that such oviposition-induced responses are known in gymnosperm and angiosperm plants (perennial and annual), this defence strategy could be either highly conserved or a strategy that has evolved convergently in different taxa [61].

Recent studies of herbaceous plants support the idea that recruiting natural enemies can increase plant fitness [81–83]. Even though folivorous insects on pine and conifers usually do not kill the plant, their feeding damage might reduce the competitiveness and fitness of the plant. Studying the costs and benefits of indirect defence in long-living trees is much more complex than in herbaceous plants because it is difficult to assess fitness gains or costs

in plants with long reproductive cycles. However, because photosynthesis activity is necessary to build up new biomass and to ensure fitness resources, measurements of net photosynthetic activity can also give an idea of the possible costs of induction. For example, net photosynthesis was found to be reduced in oviposition-induced Scots pine after egg deposition by *D. pini* [84], indicating that the cost of induction by egg deposition and recruitment of egg parasitoids by oviposition-induced pine volatiles is 'paid' by a lowered activity in primary metabolism.

Conclusions and future challenges

Like other conifers, pines have evolved effective chemical defence mechanisms against a wide range of herbivorous insects. The plant cues used by pine herbivores and their natural enemies can be constitutive as well as induced by herbivores and their associates. Many of these interactions are influenced by terpenoid and phenolic compounds from the resin and *vice versa*. However, specialized herbivorous insects have adapted to certain secondary pine metabolites, so the defensive function of these compounds is not always clear-cut. Therefore, the selection pressure on pines to alter resin chemistry to prevent further adaptation of herbivores to these components and to facilitate foraging behaviour of natural enemies is multi-dimensional. When studying induced responses in pines and other conifers they should not be restricted to biotrophic interactions but should include higher trophic levels. Compared with other ecosystems, conifer forests are fairly stable habitats and therefore it seems likely that indirect defence mechanisms, either induced by egg deposition or feeding, have evolved in several tritrophic systems convergently. Plants can influence the effectiveness of parasitoids positively and negatively, and future studies should elucidate the relative importance of these defensive traits and how they are regulated.

It would be interesting to study how herbivore-induced changes of volatiles in pine foliage are regulated at the molecular and physiological level. Recent studies indicate that jasmonates can induce similar but not identical defense response in conifers and that induced gene expression in needles differs from that in other tissues [55,56,85].

More studies are needed to elucidate the mechanisms used by herbivores and carnivores to adapt to pine terpenoids and phenolics. Little is known about how defoliators and their natural enemies detoxify these compounds [76]. Both the herbivores and carnivores in a pine forest need to cope with high levels of variability in the constitutive and herbivore-induced pine cues. Their abilities to perceive these cues and to integrate them properly are amazing considering the huge quantities of 'data' (i.e. plant cues that can be recognized and that initiate a specific behaviour). Adequate statistical approaches as well as neurobiological methods could help to understand how insects adjust their responses to the huge variety and variability of pine cues.

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